Exploring the Usage of Generative AI by The Bengaluru Urban Upper Middle Class

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***Abstract*—This research paper explores the usage patterns of generative artificial intelligence (or GenAI) tools among Indians of the urban upper middle class, residing in Bengaluru. It examines factors associated with the unified theory of acceptance and use of technology (UTAUT2) and their effect on whether or not a person is a frequent user of a GenAI tool, along with analysis of how these opinions differ across ages and genders. This paper also examines the opinion of this group regarding ethical issues with AI and its societal impact.**

Keywords—generative AI, GenAI, ChatGPT, UTAUT2, Ethical Issues with Artificial Intelligence.

I. INTRODUCTION

The usage of artificial intelligence for a range of different tasks, including content writing, code generation, research and image generation has become remarkably easier with the rise of generative AI chatbots. The speed and efficiency with which one can produce outputs for a task has greatly increased because of tools such as ChatGPT, Perplexity, DeepSeek, Gemini, Github Copilot and others. Any task that takes input in words and produces output in either words, images or other multimedia can now be automated by any person with an internet connection and a browser. In light of this, this research paper seeks to examine how such tools are used as well as what opinions people hold, with respect to the urban upper middle-class residents of Bengaluru, India.

This paper explores some of the factors connected with the unified theory of acceptance and use of technology (UTAUT2) [7][8], namely effort expectancy, social influence and performance expectancy, along with factors not connected with UTAUT2, namely trust, ease of use, ethical considerations, perceived social impact and efficiency. This paper also takes into consideration demographic factors, namely age group, gender and profession. These factors are examined against the frequency with which a person uses GenAI tools, as a measure of their adoption of the technology.

II. REVIEW OF LITERATURE

A. Bick, A. Blandin, and D. Deming [1] reports results from a series of nationally representative U.S. surveys of generative AI use at work and at home.

Huy, L., Nguyen, H., Vo-Thanh, T., Thinh, N., & Thi Thu Dung, T. [2] developed an integrated research model to explore the factors affecting ChatGPT use and its subsequent effects on whether users continue using ChatGPT and recommend it to others. They found that, first, most UTAUT2 and TTF dimensions affected ChatGPT use, and that effort expectancy, social influence, and trust had no effect on ChatGPT use.

J. O. Wecks, J. Voshaar, B. J. Plate, and J. Zimmermann [3] found that while GenAI may offer benefits for learning and engagement, the way students actually use it correlates with diminished exam outcomes.

H. Xu, R. Xu, H. Lin, and X. He [4] found that AI generated content quality significantly influences the activation of enterprise innovation behaviour, AI experience has a notable positive impact on innovation behaviour activation, and a supportive AI usage environment significantly influences the activation of enterprise innovation behaviour.

III. METHODOLOGY

A. Survey for data collection

A qualitative survey was created and sent to urban upper middle class Bengaluru residents. The respondents filled out the questionnaire online, and a total of 76 responses were received. The questionnaire obtained demographic information, namely gender, age group and profession, along with answers to the following questions:

- Do you use GenAI tools?
- Which GenAI tools do you use?
- Do you have a premium version of any GenAI tool?
- What do you primarily use GenAI for?
- When did you last use a GenAI tool?
- When did you start using GenAI?

The next section of the survey took responses to the following questions, using a 5-point Likert scale:

- GenAI saves me time. [effort expectancy]
- GenAI increases my productivity. [efficiency]
- GenAI enhances the quality of my work. [performance expectancy]
- I use GenAI tools because everybody around me uses them. [social influence]
- I am able to frame prompts easily. [ease of use]
- The environmental impact of AI influences how I use GenAI tools. [ethical consideration]
- I am aware of and cautious of potential racial, gender-based and other biases in AI. [ethical consideration]

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- I believe that AI tools negatively impact society. [perceived social impact]
- I trust AI generated responses. [trust]

B. *Analysis*

The data was pre-processed by imputing missing values, mapping the responses on the 5-point Likert scale and computing the ethical considerations factor as the mean of the two factors, environmental impact and bias in AI.

After preprocessing, exploratory data analysis was done on the data. Then, the association of the factors- effort expectancy, performance expectancy, social influence, ethical considerations, trust, perceived social impact, and ease of use- with gender was measured using independent t-tests.

Similarly, the association between the aforementioned factors and age group was measured using one-way ANOVA and Tukey's test of honest significant difference.

Furthermore, it was desired to examine the predictive power of effort expectancy, social influence and performance expectancy on frequency of GenAI usage. To solve this task, logistic regression was used.

IV. ANALYSIS AND RESULTS

The responses to the survey can be summarized as in table 1.

Factor	Value	Counts
Effort Expectancy	5.0	31
	4.0	23
	3.0	12
	0.0	7
	1.0	1
Performance Expectancy	4.0	28
	5.0	22
	3.0	14
	0.0	7
	2.0	2
	1.0	1
Social Influence	2.0	26
	3.0	18
	4.0	11
	1.0	8
	5.0	4
Efficiency	5.0	31
	4.0	23
	3.0	12
	0.0	7
	1.0	1
Ease of Use	4.0 29	29
	5.0 18	18
	3.0 17	17
	0.0 7	7
	2.0 3	3
Ethical Concerns	3.5	20
	3.0	19
	4.0	18
	4.5	7
	2.5	5
	5.0	4
	2.0	1
Trust	3.0	38
	4.0	20
	2.0	10
	1.0	4
	5.0	2
User	Yes (1)	61
	No (0)	13
Gender	Male (1)	47
	Female/Other (0)	27
Frequent Usage	Yes (1)	53
	No (0)	21

Table 1 (Counts for each factor from survey response)

Number of GenAI users by age group can be seen in table 2.

Age Group	Not GenAI User (0)		GenAI User (1)
<25	2		30
25-54	2		28
>55	9		3

Table 2 (GenAI users by age group)

The use cases of GenAI by profession can be visualised in figure 1.

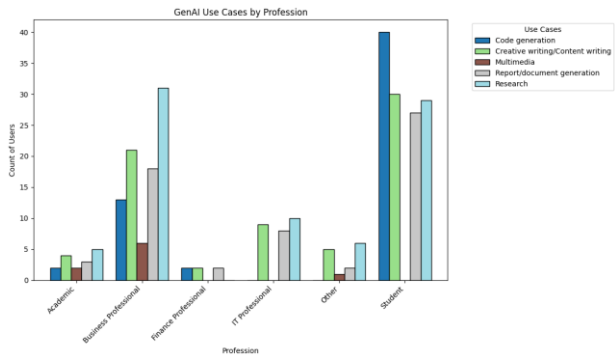

Fig 1 (GenAI Use Cases by Profession)

We observe that students primarily use AI for code generation, followed by creative/content writing, research and report/document generation. Business professionals also seem to primarily use GenAI for research, followed by creative/content writing.

Not many respondents seem to use GenAI tools for multimedia, such as image generation.

The use cases of GenAI by tool used can be visualised in figure 2

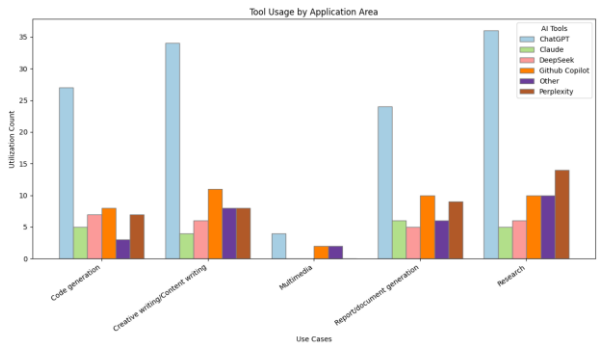

Fig 2 (Tool usage by application area)

We see that ChatGPT seems to dominate the other GenAI tools across use cases. The pattern of tools for each use case is roughly similar. Perplexity and Github Copilot seem to have a similar size of user base.

Next, independent t-test was performed for gender against the various factors:

We establish the following null hypotheses for the analysis:

H_{0PE}: There is no significant difference between mean performance expectancy between genders.

H_{0EE}: There is no significant difference between mean effort expectancy between genders.

H_{0SI}: There is no significant difference between mean social influence between genders.

H_{0EC}: There is no significant difference between mean ethical concern between genders.

H_{0T}: There is no significant difference between mean trust between genders.

H_{0PS}: There is no significant difference between mean perceived social impact between genders.

H_{0EU}: There is no significant difference between mean ease of use between genders.

Against appropriate alternative hypotheses.

Upon performing the test, the following results were obtained (as shown in table 3):

Factor	**t-statistic**	**p-value**
Performance Expectancy	2.407	0.0205
Effort Expectancy	2.396	0.0212
Social Influence	1.810	0.0767
Ethical Concerns	0.774	0.4424
Trust	2.125	0.0401
Perceived Social Impact	-1.618	0.1109
Ease of Use	0.759	0.4525

Table 3 (Gender-wise p-values for t-test)

Thus, we reject H_{0PE}, H_{0EE} and H_{0T} at 5% level of significance, concluding that mean performance expectancy, effort expectancy and trust differ across genders.

Calculating the group specific means for each factor we obtain (as shown in table 4):

Interpretation:

- There is a significant difference in mean performance expectancy, effort expectancy and trust with respect to gender.
- Males tend to trust AI generated responses more than other genders.
- All genders are neutral about ethical concerns.
- All genders lean towards agreement that AI does not negatively impact society.
- Males have higher mean effort and performance expectancy from GenAI tools.

Factor	Female/Other	Male
PE	3.074	3.957
EE	3.259	4.170
SI	2.037	2.617
EC	3.500	3.628
Trust	2.778	3.255
PSI	3.074	2.681

Table 4 (Gender-wise means of factors)

One-way ANOVA was performed with age group as the treatment factor, with yields as performance expectancy, effort expectancy, social influence, trust, ease of use, perceived social impact and ethical concerns. The following null hypotheses were used:

H_{0PE}: There is no significant difference between mean performance expectancy between age groups.

H_{0EE}: There is no significant difference between mean effort expectancy between age groups.

H_{0SI}: There is no significant difference between mean social influence between age groups.

H_{0EC}: There is no significant difference between mean ethical concern between age groups.

H_{0T}: There is no significant difference between mean trust between age groups.

H_{0PS}: There is no significant difference between mean perceived social impact between age groups.

H_{0EU}: There is no significant difference between mean ease of use between age groups.

Against appropriate alternative hypotheses.

The results of the analysis are as obtained in table 5.

Factor	F-statistic	p-value
Performance Expectancy	3.373	0.040
Effort Expectancy	2.883	0.063
Social Influence	2.264	0.111
Ethical Concerns	0.756	0.473
Trust	1.208	0.305
Perceived Social Impact	3.680	0.030
Ease of Use	4.402	0.016

Table 5 (F-statistic and p-value for ANOVA performed for age groups.)

Thus, we reject H_{0PE}, H_{0PS} and H_{0EU} at 5% level of significance, concluding that mean performance expectancy, perceived social impact and ease of use differ significantly across age groups.

Next, we use Tukey's test for honest significant difference to further examine differences in the mean effect of age group on significant responses. The results for performance expectancy, perceived social impact and ease of use are given in tables 6, 7 and 8 respectively.

Group 1	Group 2	Mean Diff.	Adj. p-value
25-54	<25	0.044	0.992
25-54	>55	-1.133	0.056
<25	>55	-1.177	0.043

Table 6 (Tukey HSD results for performance expectancy)

Group 1	Group 2	Mean Diff.	Adj. p-value
25-54	<25	0.406	0.263
25-54	>55	0.917	0.027
<25	>55	0.510	0.304

Table 7 (Tukey HSD results for perceived social impact.)

Group 1	Group 2	Mean Diff.	Adj. p-value
25-54	<25	-0.017	0.999
25-54	>55	-1.267	0.020
<25	>55	-1.250	0.021

Table 8 (Tukey HSD results for effort expectancy.)

Group-wise means were calculated as shown in fig. 3

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Age-wise Mean Scores:
           Performance Expectancy  Effort Expectancy  Social Influence  \
Age Group
25-54                    3.800000           4.033333          2.066667
<25                      3.843750           4.000000          2.750000
>55                      2.666667           2.916667          2.333333

           Ethical Concerns     Trust  Perceived Social Impact  Ease of Use
Age Group
25-54              3.516667  3.233333                 2.500000     3.766667
<25                3.687500  2.906250                 2.906250     3.750000
>55                3.458333  3.166667                 3.416667     2.500000
```

Fig 3 (Means of factors by age group)

Interpretation:

- We find that the mean attitude towards ease of use, perceived social impact and performance expectancy differ significantly across age groups.
- Respondents above 55 on average lean towards disagreement regarding ease of use of generative AI.
- Respondents across age groups tend to be neutral about ethical concerns regarding AI.
- Respondents above 55 on average tend to believe less that AI increases the quality of their work, while respondents below 55 on average lean towards agreement on this factor.
- Respondents above 55 are more likely to believe that AI negatively impacts society.

Finally, to assess the predictive power of the factors on frequency of usage of GenAI tools, a logistic regression model was built.

The factor 'Frequent User' was obtained by coding the responses to the question 'When did you last use a GenAI tool?', with levels 'Today', 'This week', 'Two weeks ago', 'One month ago' and 'More than one month ago'. The responses were coded as 'Today' and 'This week' being frequent (class 1) and others being infrequent (class 0).

Upon coding the factor, logistic regression model was built using the independent factors performance expectancy, effort expectancy, social influence, trust, ease of use, perceived social impact and ethical concerns. The results are obtained as in fig. 4.

The model has 87% accuracy, with 88% precision and 94% recall for predicting class 1 (frequent GenAI users), and 83% precision and 71% recall for predicting class 0 (infrequent users).

The coefficients obtained are as in fig. 5.

Interpretation:

- Performance expectancy, ease of use and trust have the largest weights in the logistic regression model.
- There is a negative relationship between trust and frequency of usage, indicating that despite mistrust of AI-generated responses, respondents still use the tools frequently.
- Perceived social impact also moderately negatively influences frequency of usage.

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Accuracy: 0.87
ROC-AUC: 0.88

Confusion Matrix:
[[ 5  2]
 [ 1 15]]

Classification Report:
              precision    recall  f1-score   support

           0       0.83      0.71      0.77         7
           1       0.88      0.94      0.91        16

    accuracy                           0.87        23
   macro avg       0.86      0.83      0.84        23
weighted avg       0.87      0.87      0.87        23
```

Fig. 4 (Logistic regression result scores)

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Feature Coefficients:
Performance Expectancy     1.144542
Ease of Use                0.879782
Ethical Concerns           0.138408
Effort Expectancy         -0.133063
Efficiency                -0.133063
Perceived Social Impact   -0.148611
Trust                     -0.463327
```

Fig. 5 (Coefficients of logistic regression model)

PRIMARY CONCLUSIONS

The study finds most respondents to be GenAI users, and most GenAI users to be frequent users. However, demographic factors such as gender and age group have some connection with the respondents' opinions on GenAI. Interestingly, we find that trust in AI generated responses does not have any positive effect on usage of GenAI tools. Most of the respondents seem to be uniformly neutral about the ethical concerns regarding AI. We also find that respondents above 55 years of age tend to find AI less easy to use, and also are more likely to believe that AI negatively impacts society.

These conclusions have all been made using a relatively small sample size of 74 respondents. Further work in context of this study would ideally include a much larger survey size to verify the results obtained in this paper.

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